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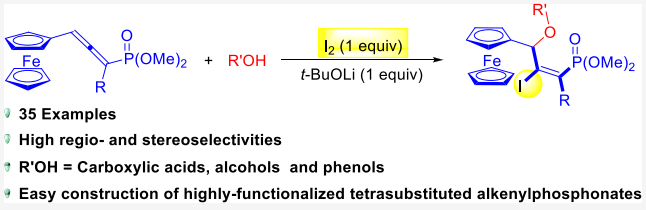
Regio- and Stereoselective Synthesis of Highly Functionalized Tetrasubstituted Olefins by Iodine-Mediated Iodofunctionalization of Ferrocene-Containing Allenylphosphonates

Le Guo, Yanpeng Gao, Zishuo Wang, Zhi Li, Yingjie Zhang, Haidong Wang, Xihui Chen, Zhenbo Gong, and Shufeng Chen*

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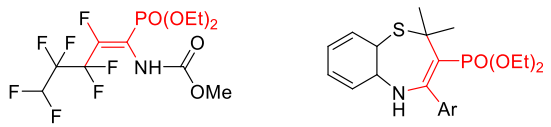
ABSTRACT: An iodine-mediated iodofunctionalization of ferrocene-containing allenylphosphonates with various carboxylic acids, alcohols, and phenols has been established, affording a series of highly functionalized tetrasubstituted alkenylphosphonate derivatives with high regio- and stereoselectivities in moderate to good yields. This approach provides an opportunity to the useful but not readily available tetrasubstituted olefins. The stereoselectivity of the reaction may be controlled by the neighboring group participation of the oxygen atom of the phosphate group.



INTRODUCTION

The alkene moiety is among the most pervasive functional groups present in various organic frameworks. In this regard, tetrasubstituted olefins are compelling structural motifs found in many natural products,[1] pharmaceutical,[2] and organic functional materials.[3] Moreover, they are also versatile intermediates for complex molecule syntheses as they are subjects of a wide range of transformations.[4] Therefore, enormous attention has been paid to their high selective and functionalized synthesis.[5] Among various approaches, the electrophilic addition of substituted allenes appears to be most efficient and modular, providing an interesting case of tetrasubstituted olefin formation.[6] However, due to the presence of two C═C bonds, controlling regioselectivity as well as stereoselectivity of the reaction to the preparation of geometrically defined tetrasubstituted olefins is still a formidable challenge. In addition, the application of the aforementioned methods to the syntheses of tetrasubstituted olefins is often limited in the mono- or disubstituted allenes. Therefore, the development of an efficient method for the construction of diverse functionalized tetrasubstituted alkenes from congested trisubstituted allenes remains highly sought after.

On the other hand, the alkenylphosphonate compound (Figure 1), which has been applied to pharmaceuticals and agrochemicals as well as various transformations to afford other functional molecules, is an attractive target.[7] Therefore, significant efforts have been devoted to the preparation of this type of compound. Traditionally, Wittig−Horner and transition-metal catalyzed Michaelis−Arbuzov reactions were the main synthetic methods for the preparation of these alkenylphosphonates with defined stereochemistry.[8] Moreover,



enzyme inhibitor antidiabetogenic drugs

Figure 1. Example of bioactive molecules containing a tetrasubstituted alkenylphosphonate skeleton.

Heck-type reactions were also successfully applied in the synthesis of alkenylphosphonates using various coupling partners.[9] However, some of these methods suffer from drawbacks such as functional group tolerance, relatively harsh reaction conditions, and the use of difficult-to-access starting materials. In addition, transition-metal-catalyzed hydrophosphorylation of alkynes has become an attractive way to synthesize vinylphosphonates,[10] but in most cases only terminal alkynes are effective. Recently, a palladium-catalyzed coupling reaction of α-diazo or *N*-tosylhydrazone phosphonates with benzyl or allyl halides for the stereoselective syntheses of trisubstituted vinylphosphonates has been reported by Wang and co-workers.[11] However, methods to synthesize diverse functionalized tetrasubstituted alkenylphosphonates are still limited.[12] We envisioned that the

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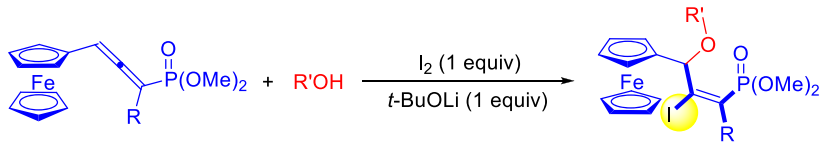


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Scheme 1. Synthesis of Ferrocene-Containing Tetrasubstituted Alkenylphosphonates

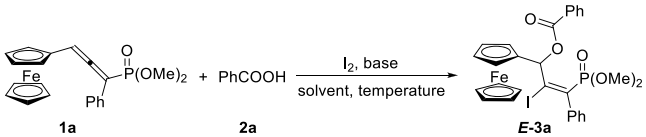


electrophilic addition of trisubstituted allenylphosphonates may provide an efficient route to the preparation of tetrasubstituted alkenylphosphonates despite the existing challenge of controlling reactivity as well as selectivity of the reaction. Therefore, in continuation of our interest in the functionalization of allenes,[13] herein we report an iodine-mediated highly regio- and stereoselective iodofunctionalization reaction of allenylphosphonates with a series of alcohols, phenols, and carboxylic acids, which provides a straightforward and convenient route to not readily available functionalized tetrasubstituted alkenylphosphonate compounds with excellent selectivity (Scheme 1).

RESULTS AND DISCUSSION

With this idea in mind, the electrophilic addition of allenylphosphonate **1a** with benzoic acid **2a** and I_2 was first chosen as the model reaction to optimize the reaction conditions (Table 1). Gratifyingly, the desired product (*E*)-**3a** was obtained in a 45% yield as the only stereoisomer when the reaction was carried out in tetrahydrofuran (THF) at room temperature (r.t.) for 1 h in the presence of 1 equiv of NaOH in an air atmosphere (Table 1, entry 1). Other solvents, such as CH_3CN, CH_3OH, toluene, and dioxane, furnished the desired product in 9, 19, 35, 42% yields, respectively (Table 1, entries 2−5). Similarly, dichloromethane (DCM) was also confirmed to be suitable for this transformation, furnishing (*E*)-**3a** in a slightly higher yield (Table 1, entry 6). Subsequently, we carefully examined the effect of the base on this electrophilic addition reaction. KOH and Cs_2CO_3 were found to be less efficient (Table 1, entries 7 and 8). The reaction with *t*-BuOLi provided (*E*)-**3a** in a high isolated yield (Table 1, entry 9). Switching *t*-BuOLi to LiOH did not increase the yield of the product (Table 1, entry 10). Only a trace amount of (*E*)-**3a** was detected when the reaction proceeded with an organic base (Table 1, entry 11). Increasing the reaction temperature led to a drop in the yield (Table 1, entry 12). Finally, the ratio of substrates was examined, and it was found that a **1a**/**2a**/iodine/base ratio of 1.0:2.0:1.0:1.0 led to the highest yield of the desired product (Table 1, entry 13). Therefore, the conditions in entry 13 (Table 1) were identified as the optimal conditions. It should be noted that the reaction proceeded with high regio- and stereoselectivities with the double bond having *E* configuration. The exact structure of (*E*)-**3a** was further unambiguously elucidated by X-ray diffraction analysis of the single crystal.[14]

Table 1. Optimization of Reaction Conditions[a]



1a **2a** ***E*-3a**

entry	solvent	base	*T* (°C)	yield (%)[b]
1	THF	NaOH	r.t.	45
2	CH_3CN	NaOH	r.t.	9
3	CH_3OH	NaOH	r.t.	19
4	toluene	NaOH	r.t.	35
5	dioxane	NaOH	r.t.	42
6	CH_2Cl_2	NaOH	r.t.	47
7	CH_2Cl_2	KOH	r.t.	17
8	CH_2Cl_2	Cs_2CO_3	r.t.	39
9	CH_2Cl_2	*t*-BuOLi	r.t.	70
10	CH_2Cl_2	LiOH	r.t.	60
11	CH_2Cl_2	DBU	r.t.	trace
12	CH_2Cl_2	*t*-BuOLi	40	54
13[c]	CH_2Cl_2	*t*-BuOLi	r.t.	78

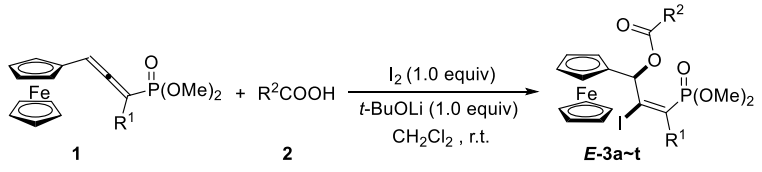
[a]The reaction was carried out using 0.20 mmol of **1a**, 0.10 mmol of **2a**, 0.15 mmol of iodine, and 0.10 mmol of base in 2.0 mL of solvent. [b]Yield of the isolated product purified by a column chromatography on silica gel. [c]**1a**/**2a**/iodine/base = 1.0:2.0:1.0:1.0; **1a** (0.10 mmol).

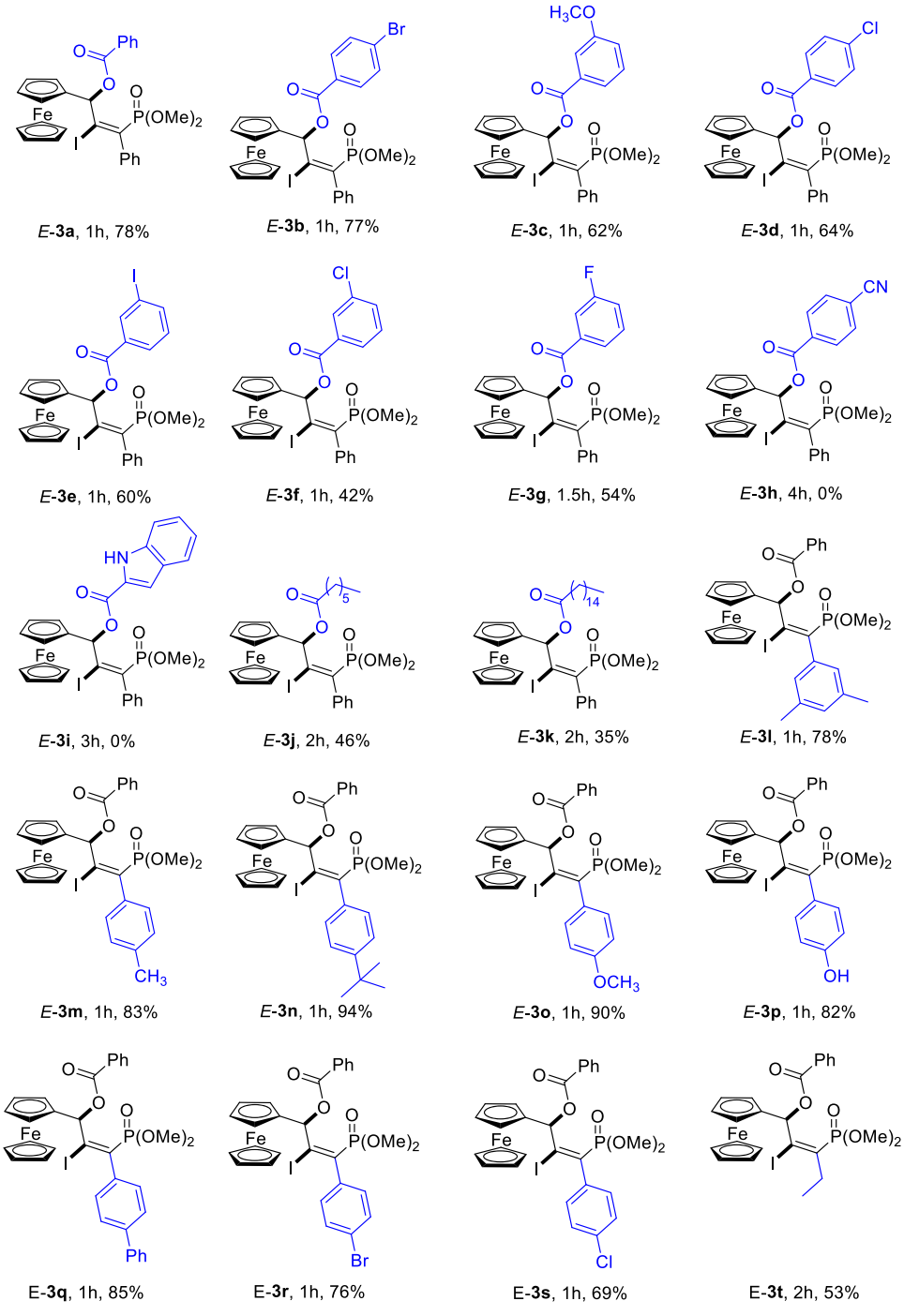
With the optimized reaction conditions in hand, we then proceeded to study the scope of the current iodofunctionalization reaction using various allenylphosphonates and carboxylic acids as the substrates, and the experimental data are shown in Scheme 2. Most aromatic acids worked well with allenylphosphonate **1a** to give ferrocene-containing tetrasubstituted alkenylphosphonates **3a**−**g** in moderate to good yields. It was found that various substituents on the phenyl ring of benzoic acids, such as fluoro, chloro, bromo, iodo, and methoxy, were well-tolerated under optimal conditions. Unfortunately, the electron-withdrawing group with 4-CN attached to the phenyl ring was proved to be fruitless, probably due to the strong electronic effect (**3h**). Moreover, heteroaryl species such as *β*-indoleacetic acid as substrate failed to afford any corresponding products (**3i**). Furthermore, two aliphatic acids were also tested and gave the corresponding products **3j** and **3k** in 46 and 35% yields, respectively. Our next objective was to check the scope of trisubstituted allenylphosphonate substrates. Delightfully, with regard to the R^1 group of the allenes, different substituents on the phenyl ring were all suitable for this transformation, affording the tetrasubstituted alkenylphosphonate products in good to high yields (**3l**−**s**). Notably, a free hydroxyl group could survive well and the reaction proceeded smoothly to give alkenylphosphonate product **3p** in an 82% yield. Moreover, except for aromatic ring groups, the alkyl substituent could also be introduced to the olefin product, albeit with a slightly lower yield (**3t**). Finally, a gram-scale experiment of **1a** with **2a** was carried out to examine the scalability, and a 66% yield of **3a** was obtained on the 2.5 mmol scale under the standard conditions, validating that this method is scalable.

The potential impact of this transformation would be expanded if alcohols could be employed as nucleophilic reagents for the reaction. Thus, to expand the scope of the tetrasubstituted alkenylphosphonate formation reaction further, diverse alcohols were employed as substrates under the

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Scheme 2. Iodine-Mediated Iodofunctionalization of Various Allenylphosphonates and Carboxylic Acids[a]



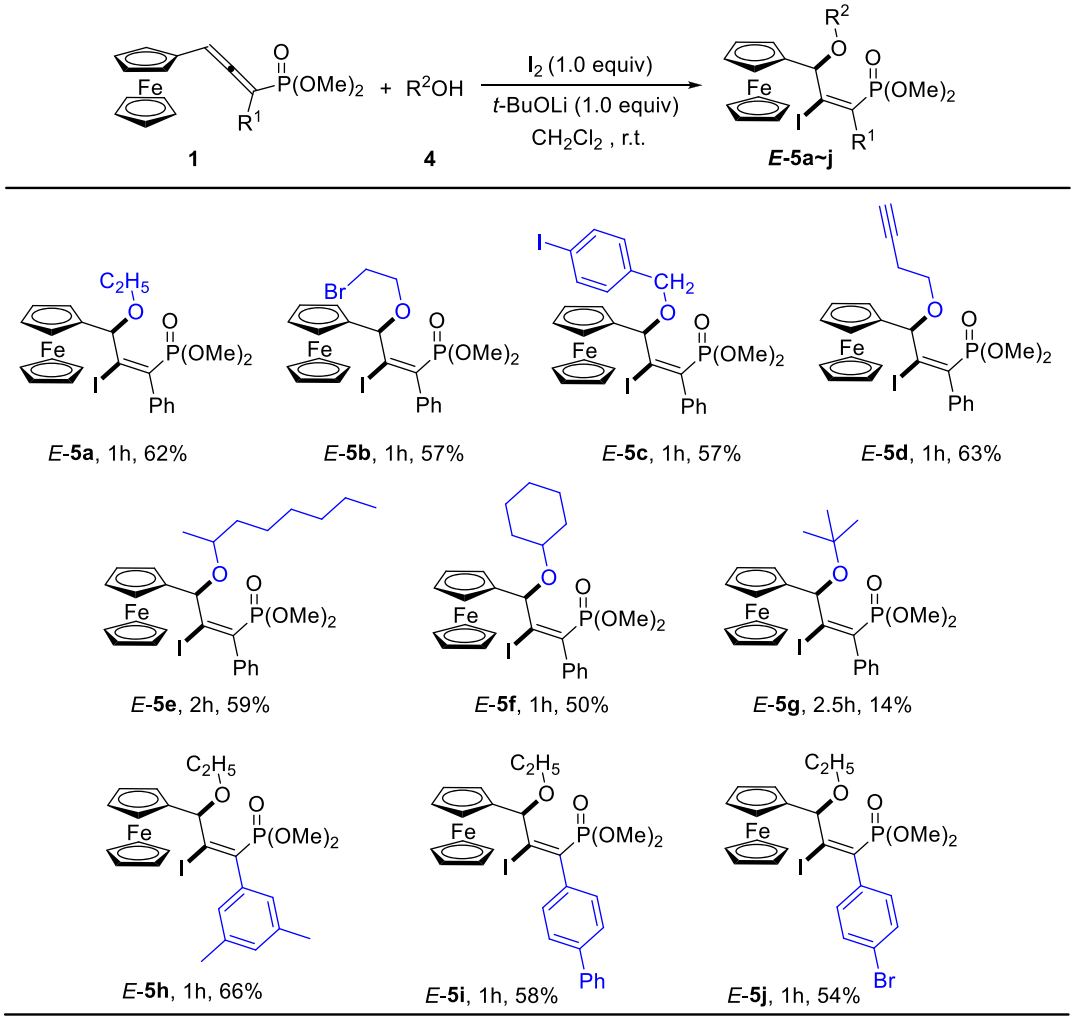


[a]Reactions were carried out with 0.20 mmol of **1**, 0.40 mmol of **2**, 0.20 mmol of iodine, and 0.20 mmol of *t*-BuOLi in 3.0 mL of DCM at room temperature. Yield of the isolated product after chromatography.

same reaction conditions (Scheme 3). We were pleased to find that ethanol reacted successfully with **1** to provide only *E*-isomer product **5a** in a 62% yield. Similarly, other primary alcohols bearing various functional groups, such as 2-bromoethanol, (4-iodophenyl) methanol, and but-3-yn-1-ol, proved to be good substrates, providing the corresponding

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Scheme 3. Highly Regio- and Stereoselective Iodoalkoxylation of Allenylphosphonates with Various Alcohols[a]



[a]Reactions were carried out with 0.20 mmol of **1**, 0.40 mmol of **4**, 0.20 mmol of iodine, 0.20 mmol of *t*-BuOLi in 3.0 mL of DCM at room temperature. Yields of the isolated product after chromatography.

products with satisfactory yields (**5b**−**d**). Moreover, second alcohols were also suitable for this reaction, affording the desired products **5e** and **5f** in 59 and 50% yields, respectively. However, increasing the steric hindrance further by employing a tertiary alcohol resulted in the alkenylphosphonate product **5g** being obtained in a lower yield, probably because of the steric hindrance effect. Finally, randomly selected three trisubstituted allenylphosphonate compounds were subjected to the reaction and furnished the corresponding products in moderate to good yields as a single stereoisomer in each case (**5h**−**j**).

Encouraged by the success in the application of alcohols, we next turned our attention to expanding the substrate scope to phenols. We found that under optimal reaction conditions, as shown in Scheme 4, the iodine-mediated iodofunctionalization of allenylphosphonate with phenols led to the formation of *E*-tetrasubstituted alkenylphosphonates in moderate yields with complete stereoselectivity. When an electron-donating substituent such as methyl group was introduced, the reaction proceeded smoothly (**7b**), while *tert*-butyl group decreased the reaction efficiency (**7c**). In contrast, the desired product was not formed when a strong electron-withdrawing group was incorporated (**7d**). Moreover, halogen substituted phenols were all efficiently coupled to furnish the corresponding products in moderate yields (**7e**−**g**). However, the ortho substituent hampered the reaction and no desired product was detected (**7h**). Finally, the reaction of 4-phenylphenol gave rise to **7i** in a 40% isolated yield.

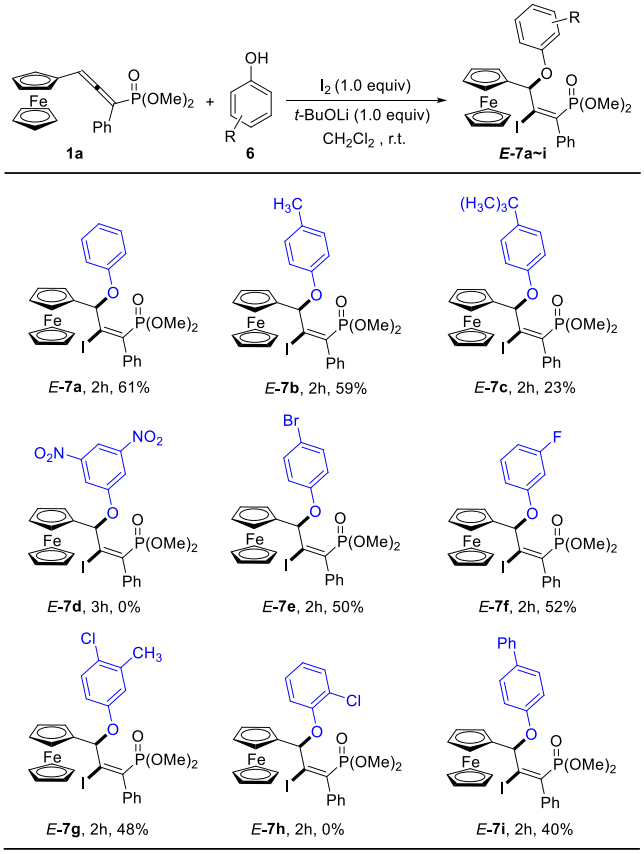
The high regio- and stereoselectivity for ferrocene-containing tetrasubstituted alkenylphosphonate formation may be interpreted by the protocol shown in Scheme 5. Initially, the iodonium intermediate **A** was formed regioselectively by the direct electrophilic addition of the relatively electron-rich carbon−carbon double bond with I_2. Next, a five-membered intermediate **B** was generated via the neighboring group participation of the oxygen atom of the phosphate group.[12b] Finally, the oxygen nucleophile attacks at the α position of the ferrocene group accompanied by cleavage of the C−O bond, which afforded the highly stereoselective *E*-isomer as the final product.

■ CONCLUSIONS

We have developed a practical method for the construction of useful tetrasubstituted alkenylphosphonates via iodine-mediated electrophilic addition of ferrocene-containing allenylphosphonates with various oxygen nucleophiles in high regio-

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Scheme 4. Synthesis of Tetrasubstituted Alkenylphosphonates by the Reaction of Allenylphosphonate 1a with Various Phenols[a]



[a]Reactions were carried out with 0.20 mmol of **1a**, 0.40 mmol of **6**, 0.2 mmol of iodine, 0.20 mmol of *t*-BuOLi in 3.0 mL of DCM at room temperature. Yield of the isolated product after chromatography.

and stereoselectivity. The approach provided a series of highly functionalized tetrasubstituted olefins in moderate to excellent yields with good functional tolerance. It is believed that the stereoselectivity of the reaction may be controlled by the neighboring group participation of the oxygen atom of the phosphate group. We are confident that this protocol would allow the synthesis of many highly substituted allenylphosphonate analogues.

EXPERIMENTAL SECTION

General Information. All commercially available reagents were used directly without purification unless otherwise stated. All solvents were purified following standard procedures. For chromatography, 100−200 mesh silica gel (Qingdao, China) was employed. ^{1}H and ^{13}C NMR spectra were recorded at 500 and 125 MHz, respectively. Chemical shifts are reported in ppm using tetramethylsilane as internal standard when $CDCl_3$ was used as solvent. IR spectra were recorded on a Fourier transform infrared (FT-IR) instrument. The high-resolution mass spectrometry (HRMS) analysis was obtained on a quadrupole time-of-flight (QTOF) mass spectrometer. Melting points were determined with melting point apparatus and are uncorrected. Allenylphosphonates **1** were prepared according to our previously described procedures.[15]

General Procedure for the Electrophilic Addition of Ferrocene-Containing Allenylphosphonates with Carboxylic Acids. To a solution of allenylphosphonate **1** (0.20 mmol), carboxylic acids **2** (0.40 mmol), and *t*-BuOLi (0.20 mmol) in CH_2Cl_2 (3.0 mL) was added I_2 (0.20 mmol) under a N_2 atmosphere. The resulting mixture was stirred at room temperature for the indicated time. The solvent was removed in a vacuum, and the resulting residue was purified on a silica gel column (petroleum ether/EtOAc = 5:1) to offer yellow solid products (*E*)-**3**.

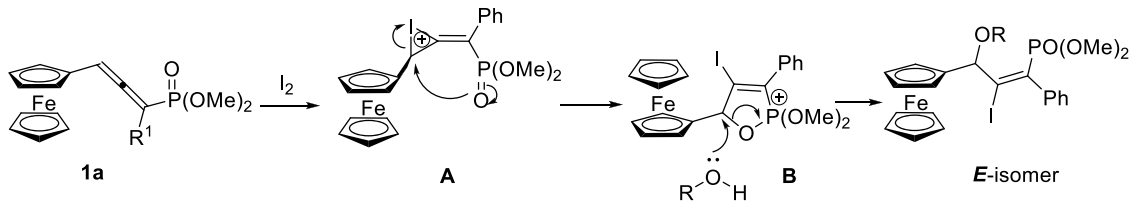
*Large-Scale Preparation of **3a**.* To a solution of allenylphosphonate **1a** (2.5 mmol, 1.02 g), benzoic acid **2a** (5.0 mmol, 0.61 g), and *t*-BuOLi (2.5 mmol, 0.20 g) in CH_2Cl_2 (25.0 mL) was added I_2 (2.5 mmol, 0.63 g) under a N_2 atmosphere. The resulting mixture was stirred at room temperature for 2 h. The solvent was removed in a vacuum, and the resulting residue was purified on a silica gel column (petroleum ether/EtOAc = 5:1) to offer a yellow solid product (*E*)-**3a** (1.08 g, 66%).

*(E)-Dimethyl(3-ferrocenyl-3-phenylmethylester-2-iodo-1-phenylpropenyl)phosphonate ((E)-**3a**).* Light yellow solid; 100 mg, 78% yield; R_f = 0.43 (petroleum ether/ethyl acetate = 5:1); mp 150−151 °C. IR (neat, cm^{-1}) 2922, 2852, 1719, 1574, 1454, 1261, 1106, 1027, 814, 762, 710. ^{1}H NMR (500 MHz, $CDCl_3$) δ 8.25 (d, J = 7.3 Hz, 2H), 8.13 (d, J = 7.5 Hz, 1H), 7.69 (d, J = 2.5 Hz, 1H), 7.52 (t, J = 7.7 Hz, 2H), 7.32 (d, J = 9.2 Hz, 3H), 7.08 (d, J = 6.8 Hz, 1H), 6.98 (s, 1H), 4.80 (s, 1H), 4.67 (s, 1H), 4.23 (d, J = 6.1 Hz, 2H), 4.19 (s, 5H), 4.00 (d, J = 11.4 Hz, 3H), 3.64 (d, J = 11.2 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 164.6, 143.0, 137.3, 137.1, 136.9, 133.4 (d, J_{pc} = 18.0 Hz), 130.2, 129.3 (d, J_{pc} = 173.8 Hz), 129.1, 128.4 (d, J_{pc} = 14.9 Hz), 128.1, 86.9, 73.2 (d, J_{pc} = 4.5 Hz), 69.2, 68.4, 68.1, 67.8, 67.2, 53.6 (d, J_{pc} = 5.2 Hz), 53.1 (d, J_{pc} = 6.1 Hz). HRMS (electrospray ionization (ESI)) m/z: [M]$^+$ calcd for $C_{28}H_{26}FeIO_5P$ 655.9907; found 655.9878.

*(E)-Dimethyl[3-ferrocenyl-3-(4-bromophenylmethylester)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**3b**).* Pale yellow oil; 110 mg, 77% yield; R_f = 0.5 (petroleum ether/ethyl acetate = 5:1); mp 152 °C. IR (neat, cm^{-1}) 2921, 2851, 1723, 1587, 1557, 1482, 1263, 1102, 1025, 816, 736, 702. ^{1}H NMR (500 MHz, $CDCl_3$) δ 8.13 (d, J = 8.5 Hz, 2H), 7.71 (dd, J = 10.0, 5.6 Hz, 3H), 7.35 (s, 3H), 7.09 (d, J = 6.5 Hz, 1H), 7.00 (s, 1H), 4.81 (s, 1H), 4.65 (s, 1H), 4.26 (d, J = 6.9 Hz, 2H), 4.21 (s, 5H), 4.00 (d, J = 11.4 Hz, 3H), 3.66 (d, J = 11.2 Hz, 3H). ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$) δ 163.9, 143.0, 132.0, 131.5, 129.1, 128.5, 128.5, 128.3, 128.1, 86.7, 73.4 (d, J_{pc} = 4.4 Hz), 69.2 (d, J_{pc} = 19.2 Hz), 68.4, 68.1, 67.8, 67.1, 53.5 (d, J_{pc} = 5.2 Hz), 53.0 (d, J_{pc} = 6.2 Hz). HRMS (ESI) m/z: [M]$^+$ calcd for $C_{28}H_{25}BrFeIO_5P$ 733.9013; found 733.9014.

*(E)-Dimethyl[3-ferrocenyl-3-(3-methoxyphenylmethylester)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**3c**).* Light yellow solid; 83 mg, 62% yield; R_f = 0.45 (petroleum ether/ethyl acetate = 5:1); mp 157−158 °C. IR (neat, cm^{-1}) 2921, 2852, 1721, 1645, 1584, 1462, 1274, 1032, 818, 754, 701. ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.86 (d, J = 6.9 Hz, 1H), 7.77 (s, 1H), 7.68 (s, 1H), 7.44 (t, J = 8.0 Hz, 1H),

Scheme 5. Rationalization of Regio- and Stereoselectivity



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7.33 (d, J = 9.5 Hz, 3H), 7.17 (dd, J = 8.2, 2.5 Hz, 1H), 7.08 (d, J = 6.9 Hz, 1H), 6.98 (s, 1H), 4.79 (s, 1H), 4.65 (s, 1H), 4.23 (d, J = 7.2 Hz, 2H), 4.20 (d, J = 0.5 Hz, 5H), 3.98 (d, J = 11.4 Hz, 3H), 3.90 (s, 3H), 3.63 (t, J = 7.7 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 164.5, 159.7, 143.1, 137.1, 131.5, 129.1, 128.9 (d, J_{pc} = 195.0 Hz), 128.4 (d, J_{pc} = 26.9 Hz), 122.4, 119.7, 114.6, 86.9, 73.2 (d, J_{pc} = 4.6 Hz), 69.2, 68.3, 68.1, 67.8, 67.2, 55.5, 53.5 (d, J_{pc} = 5.3 Hz), 53.0 (d, J_{pc} = 6.2 Hz). HRMS (ESI) m/z: $[M + K]^+$ calcd for $C_{29}H_{28}FeIO_6PK$ 724.9650; found 724.9645.

*(E)-Dimethyl[3-ferrocenyl-3-(4-chlorophenylmethylester)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**3d**).* Light yellow solid; 87 mg, 64% yield; R_f = 0.5 (petroleum ether/ethyl acetate = 5:1); mp 159−160 °C. IR (neat, cm^{-1}) 2921, 2851, 1722, 1643, 1591, 1406, 1263, 1098, 1025, 817, 760, 699. ^1H NMR (500 MHz, $CDCl_3$) δ 8.20 (d, J = 8.6 Hz, 2H), 7.72 (d, J = 2.7 Hz, 1H), 7.56−7.50 (m, 2H), 7.35 (s, 3H), 7.10 (s, 1H), 7.00 (s, 1H), 4.82−4.79 (m, 1H), 4.67−4.64 (m, 1H), 4.26 (dd, J = 8.3, 1.3 Hz, 2H), 4.21 (s, 5H), 4.00 (d, J = 11.4 Hz, 3H), 3.66 (d, J = 11.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 163.8, 142.9 (d, J_{pc} = 6.4 Hz), 139.9, 138.5, 137.2, 136.8, 131.4, 129.0, 128.4 (dd, J_{pc} = 33.9, 20.0 Hz), 86.8, 73.4 (d, J_{pc} = 4.5 Hz), 69.2, 68.5, 68.1, 67.9, 67.1, 53.5 (d, J_{pc} = 5.2 Hz), 53.1 (d, J_{pc} = 6.2 Hz). HRMS (ESI) m/z: $[M]^+$ calcd for $C_{28}H_{25}ClFeIO_5P$ 689.9517; found 689.9498.

*(E)-Dimethyl[3-ferrocenyl-3-(3-iododiphenylmethylester)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**3e**).* Light yellow solid; 92 mg, 60% yield; R_f = 0.5 (petroleum ether/ethyl acetate = 5:1); mp 136−137 °C. IR (neat, cm^{-1}) 2921, 2852, 1721, 1647, 1557, 1463, 1254, 1132, 1112, 1032, 815, 753, 700. ^1H NMR (500 MHz, $CDCl_3$) δ 8.60 (s, 1H), 8.23 (d, J = 7.8 Hz, 1H), 7.98 (d, J = 7.9 Hz, 1H), 7.71 (d, J = 2.7 Hz, 1H), 7.33 (dd, J = 25.8, 8.9 Hz, 4H), 7.10 (d, J = 6.7 Hz, 1H), 7.00 (s, 1H), 4.81 (d, J = 1.1 Hz, 1H), 4.67 (s, 1H), 4.27 (dd, J = 11.6, 1.2 Hz, 2H), 4.22 (s, 5H), 4.00 (d, J = 11.4 Hz, 3H), 3.66 (d, J = 11.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 163.2, 142.9, 142.2, 138.9, 136.7, 132.0, 129.7 (d, J_{pc} = 156.3 Hz), 128.4, 128.2, 94.1, 86.6, 73.7 (d, J_{pc} = 4.5 Hz), 69.2, 68.4, 68.1, 67.9, 67.1, 53.6 (d, J_{pc} = 5.2 Hz), 53.1 (d, J_{pc} = 6.1 Hz). HRMS (ESI) m/z: $[M]^+$ calcd for $C_{28}H_{25}FeI_2O_5P$ 781.8873; found 781.8861.

*(E)-Dimethyl[3-ferrocenyl-3-(3-chlorophenylmethylester)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**3f**).* Light yellow solid; 57 mg, 42% yield; R_f = 0.5 (petroleum ether/ethyl acetate = 5:1); mp 138−139 °C. IR (neat, cm^{-1}) 2921, 2851, 1725, 1575, 1465, 1424, 1250, 1123, 1027, 813, 742, 699. ^1H NMR (500 MHz, $CDCl_3$) δ 8.21 (t, J = 1.7 Hz, 1H), 8.13 (d, J = 7.8 Hz, 1H), 7.70 (d, J = 2.6 Hz, 1H), 7.60 (dd, J = 8.0, 1.0 Hz, 1H), 7.47 (t, J = 7.9 Hz, 1H), 7.33 (d, J = 9.6 Hz, 3H), 7.07 (d, J = 7.0 Hz, 1H), 6.98 (s, 1H), 4.79 (s, 1H), 4.65 (s, 1H), 4.25 (dd, J = 11.1, 1.1 Hz, 2H), 4.19 (s, 5H), 3.98 (d, J = 11.4 Hz, 3H), 3.64 (d, J = 11.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 163.5, 142.9, 138.7, 136.4, 134.8, 132.7 (d, J_{pc} = 183.8 Hz), 130.0 (d, J_{pc} = 10.1 Hz), 128.1 (d, J_{pc} = 8.0 Hz), 86.6, 73.7, 69.2, 68.4, 68.1, 67.9, 67.1, 53.5 (d, J_{pc} = 5.2 Hz), 53.0 (d, J_{pc} = 6.1 Hz). HRMS (ESI) m/z: $[M]^+$ calcd for $C_{28}H_{25}ClFeIO_5P$ 689.9517, found 689.9526.

*(E)-Dimethyl[3-ferrocenyl-3-(3-fluorophenylmethylester)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**3g**).* Light yellow solid; 75 mg, 54% yield; R_f = 0.53 (petroleum ether/ethyl acetate = 5:1); mp 142−143 °C. IR (neat, cm^{-1}) 2921, 2852, 1725, 1587, 1446, 1271, 1199, 1027, 823, 754, 701. ^1H NMR (500 MHz, $CDCl_3$) δ 8.04 (d, J = 7.7 Hz, 1H), 7.92 (d, J = 9.1 Hz, 1H), 7.71 (d, J = 2.5 Hz, 1H), 7.51 (td, J = 8.0, 5.7 Hz, 1H), 7.33 (dd, J = 7.7, 5.3 Hz, 4H), 7.07 (d, J = 6.8 Hz, 1H), 6.98 (s, 1H), 4.79 (s, 1H), 4.65 (s, 1H), 4.26−4.22 (m, 2H), 4.19 (s, 5H), 3.98 (d, J = 11.4 Hz, 3H), 3.63 (d, J = 11.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 163.6 (d, J_{pc} = 18.2 Hz), 161.7, 143.0 (d, J_{pc} = 6.5 Hz), 138.7, 137.3, 136.6, 136.4, 132.3 (d, J_{pc} = 7.5 Hz), 130.3 (d, J_{pc} = 7.7 Hz), 129.0, 128.4 (d, J_{pc} = 13.5 Hz), 128.1, 125.7 (d, J_{pc} = 2.9 Hz), 120.5, 120.4, 117.0, 116.8, 86.7, 73.6 (d, J_{pc} = 4.5 Hz), 69.2, 68.4, 68.1, 67.9, 67.1, 53.5 (d, J_{pc} = 5.2 Hz), 53.0 (d, J_{pc} = 6.2 Hz). HRMS (ESI) m/z: $[M + K]^+$ calcd for $C_{28}H_{25}FFeIO_5PK$ 712.9450; found 712.9430.

*(E)-Dimethyl[3-ferrocenyl-3-hexanoicmethylester-2-iodo-1-phenylpropenyl]phosphonate ((E)-**3j**).* Light yellow solid; 60 mg, 46% yield; R_f = 0.38 (petroleum ether/ethyl acetate = 5:1); mp 100−102 °C. IR (neat, cm^{-1}) 2926, 2856, 1736, 1457, 1248, 1164, 1028, 812, 699. ^1H NMR (500 MHz, $CDCl_3$) δ 7.41 (s, 1H), 7.35 (d, J = 9.0 Hz, 3H), 7.08 (d, J = 6.8 Hz, 1H), 6.98 (s, 1H), 4.73 (s, 1H), 4.54 (s, 1H), 4.24 (s, 5H), 4.22 (d, J = 5.9 Hz, 2H), 3.95 (d, J = 11.4 Hz, 3H), 3.63 (d, J = 11.1 Hz, 3H), 2.52 (t, J = 7.5 Hz, 2H), 1.84−1.77 (m, 2H), 1.39−1.26 (m, 7H), 0.93 (d, J = 6.8 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 172.0, 143.1 (d, J_{pc} = 6.5 Hz), 137.7, 137.6, 137.5 (d, J_{pc} = 163.8 Hz), 129.1, 128.4 (d, J_{pc} = 26.3 Hz), 128.0, 86.8, 72.8 (d, J_{pc} = 4.6 Hz), 69.1, 68.3, 67.9, 67.7, 67.2, 53.4 (d, J_{pc} = 5.1 Hz), 52.9 (d, J_{pc} = 6.1 Hz), 34.8, 31.5, 29.0, 25.0, 22.6, 14.1. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{28}H_{35}FeIO_5P$ 665.0611; found 665.0608.

*(E)-Dimethyl[3-ferrocenyl-3-n-pentadecanoicmethylester-2-iodo-1-phenylpropenyl]phosphonate ((E)-**3k**).* Pale yellow oil; 54 mg, 35% yield; R_f = 0.39 (petroleum ether/ethyl acetate = 5:1). IR (neat, cm^{-1}) 2922, 2853, 1739, 1646, 1248, 1178, 1031, 815, 700. ^1H NMR (500 MHz, $CDCl_3$) δ 7.40 (d, J = 2.2 Hz, 1H), 7.34 (s, 3H), 7.08 (d, J = 6.8 Hz, 1H), 6.98 (d, J = 3.9 Hz, 1H), 4.72 (s, 1H), 4.53 (s, 1H), 4.24 (s, 5H), 4.22 (s, 2H), 3.95 (d, J = 11.4 Hz, 3H), 3.63 (d, J = 11.2 Hz, 3H), 2.52 (s, 2H), 1.80 (d, J = 7.5 Hz, 2H), 1.28 (s, 22H), 0.90 (s, 6H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 172.0, 143.1 (d, J_{pc} = 6.4 Hz), 137.8, 137.6, 137.4 (d, J_{pc} = 165.0 Hz), 129.1, 128.3, 128.0, 86.7, 72.8 (d, J_{pc} = 4.5 Hz), 69.1, 68.3, 67.9, 67.7, 67.2, 53.4 (d, J_{pc} = 5.1 Hz), 53.0 (d, J_{pc} = 6.1 Hz), 34.8, 31.9, 29.7, 29.4 (d, J_{pc} = 3.2 Hz), 25.1, 22.7, 14.1. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{37}H_{52}FeIO_5P$ 790.1942; found 790.1916.

*(E)-Dimethyl[3-ferrocenyl-3-phenylmethylester-2-iodo-1-(3,5-dimethylphenyl)propenyl]phosphonate ((E)-**3l**).* Light yellow solid; 98 mg, 78% yield; R_f = 0.43 (petroleum ether/ethyl acetate = 5:1); mp 172−173 °C.IR (neat, cm^{-1}) 3374, 2921, 2852, 1722, 1591, 1263, 1098, 1025, 817, 760, 699. ^1H NMR (500 MHz, $CDCl_3$) δ 8.26 (d, J = 7.3 Hz, 2H), 7.68 (d, J = 1.9 Hz, 1H), 7.62 (t, J = 7.4 Hz, 1H), 7.52 (t, J = 7.7 Hz, 2H), 6.93 (s, 1H), 6.68 (s, 1H), 6.58 (s, 1H), 4.80 (s, 1H), 4.67 (s, 1H), 4.24 (d, J = 6.3 Hz, 2H), 4.19 (s, 5H), 3.98 (d, J = 11.4 Hz, 3H), 3.65 (d, J = 11.2 Hz, 3H), 2.26 (d, J = 13.0 Hz, 6H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 164.6, 142.8, 137.8, 133.3, 130.2, 129.8, 129.3 (d, J_{pc} = 176.3 Hz), 126.6, 126.1, 87.1, 73.2, 69.2, 68.4, 68.1, 67.8, 67.2, 53.5 (d, J_{pc} = 5.1 Hz), 53.1 (d, J_{pc} = 6.2 Hz), 21.3 (d, J_{pc} = 6.8 Hz). HRMS (ESI) m/z: $[M]^+$ calcd for $C_{30}H_{30}FeIO_5P$ 684.0220; found 684.0211.

*(E)-Dimethyl[3-ferrocenyl-3-phenylmethylester-2-iodo-1-(4-methylphenyl)propenyl]phosphonate ((E)-**3m**).* Light yellow solid; 105 mg, 83% yield; R_f = 0.39 (petroleum ether/ethyl acetate = 5:1); mp 158−159 °C. IR (neat, cm^{-1}) 3374, 2921, 2852, 1721, 1648, 1510, 1458, 1262, 1106, 1027, 828, 755. ^1H NMR (500 MHz, $CDCl_3$) δ 8.28 (d, J = 7.3 Hz, 2H), 7.73 (d, J = 2.7 Hz, 1H), 7.65 (t, J = 7.4 Hz, 1H), 7.55 (t, J = 7.7 Hz, 2H), 7.16 (s, 2H), 6.99 (d, J = 7.3 Hz, 1H), 6.90 (d, J = 7.3 Hz, 1H), 4.82 (d, J = 1.0 Hz, 1H), 4.69 (s, 1H), 4.26 (dd, J = 9.6, 1.2 Hz, 2H), 4.22 (s, 5H), 4.00 (d, J = 11.4 Hz, 3H), 3.67 (d, J = 11.2 Hz, 3H), 2.36 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 164.6, 140.1 (d, J = 6.4 Hz), 138.3, 137.9, 137.1, 130.2, 129.3 (d, J_{pc} = 176.3 Hz), 129.0 (d, J_{pc} = 20.4 Hz), 128.3, 87.0, 73.2 (d, J_{pc} = 4.5 Hz), 69.1, 68.3, 68.1, 67.8, 67.2, 53.4 (d, J_{pc} = 5.1 Hz), 53.0 (d, J_{pc} = 6.1 Hz), 21.4. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{29}H_{28}FeIO_5P$ 670.0063; found 670.0038.

*(E)-Dimethyl[3-ferrocenyl-3-phenylmethylester-2-iodo-1-(4-tert-butylphenyl)propenyl]phosphonate ((E)-**3n**).* Light yellow solid; 115 mg, 94% yield; R_f = 0.45 (petroleum ether/ethyl acetate = 5:1); mp 93−94 °C. IR (neat, cm^{-1}) 2956, 2921, 2853, 1721, 1642, 1462, 1261, 1107, 1026, 831, 712. ^1H NMR (500 MHz, $CDCl_3$) δ 8.30−8.25 (m, 2H), 7.73 (d, J = 2.7 Hz, 1H), 7.65 (t, J = 7.4 Hz, 1H), 7.55 (t, J = 7.7 Hz, 2H), 7.34 (s, 2H), 7.00 (s, 1H), 6.92 (s, 1H), 4.83−4.80 (m, 1H), 4.71−4.67 (m, 1H), 4.25 (ddd, J = 4.6, 3.6, 1.8 Hz, 2H), 4.22 (s, 5H), 4.01 (d, J = 11.4 Hz, 3H), 3.65 (d, J = 11.2 Hz, 3H), 1.32 (s, 9H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 164.6, 151.0 (d, J_{pc} = 1.9 Hz), 140.0 (d, J_{pc} = 6.5 Hz), 138.4, 137.1, 136.9, 136.8, 133.3, 130.2, 129.3 (d, J = 177.5 Hz), 125.1, 87.1 (d, J_{pc} = 1.7 Hz), 73.2 (d, J_{pc} = 4.5 Hz), 69.1, 68.3, 68.1, 67.7, 67.2, 53.4 (d, J_{pc} = 5.1

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Hz), 53.0 (d, J_{pc} = 6.0 Hz), 34.6, 31.3. HRMS (ESI) m/z: calcd for $[M]^+$ $C_{32}H_{34}FeIO_5P$ 712.0530; found 712.0504.

*(E)-Dimethyl[3-ferrocenyl-3-phenylmethylester-2-iodo-1-(4-methoxyphenyl)propenyl]phosphonate ((E)-**3o**).* Light yellow solid; 113 mg, 90% yield; R_f = 0.48 (petroleum ether/ethyl acetate = 5:1); mp 154−155 °C. IR (neat, cm^{-1}) 2921, 2851, 1721, 1641, 1506, 1460, 1251, 1107, 1029, 831, 713, 648. ^1H NMR (500 MHz, $CDCl_3$) δ 8.29 (d, J = 7.4 Hz, 2H), 7.73 (d, J = 2.2 Hz, 1H), 7.65 (t, J = 7.4 Hz, 1H), 7.55 (t, J = 7.7 Hz, 2H), 7.02 (s, 1H), 6.93 (s, 1H), 6.87 (s, 2H), 4.82 (s, 1H), 4.70 (s, 1H), 4.26 (d, J = 8.4 Hz, 2H), 4.22 (s, 5H), 4.01 (d, J = 11.4 Hz, 3H), 3.81 (s, 3H), 3.68 (d, J = 11.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 164.6, 159.3, 137.8 (d, J_{pc} = 14.7 Hz), 137.6, 136.6, 135.4 (d, J_{pc} = 6.5 Hz), 133.3, 130.2, 129.3 (d, J_{pc} = 173.8 Hz), 87.0, 73.2 (d, J_{pc} = 4.3 Hz), 69.2, 68.3, 68.1, 67.8, 67.2, 55.2, 53.5 (d, J_{pc} = 5.1 Hz), 53.0 (d, J_{pc} = 6.1 Hz). HRMS (ESI) m/z: $[M]^+$ calcd for $C_{29}H_{28}FeIO_6P$ 686.0012; found 685.9982.

*(E)-Dimethyl[3-ferrocenyl-3-phenylmethylester-2-iodo-1-(4-phenol)propenyl]phosphonate ((E)-**3p**).* Light yellow solid; 104 mg, 82% yield; R_f = 0.43 (petroleum ether/ethyl acetate = 5:1); mp 159−160 °C. IR (neat, cm^{-1}) 3360, 2921, 2852, 1721, 1645, 1262, 1027, 828, 751. ^1H NMR (500 MHz, $CDCl_3$) δ 8.29−8.24 (m, 2H), 7.70−7.63 (m, 2H), 7.56 (t, J = 7.7 Hz, 2H), 7.32 (t, J = 9.5 Hz, 2H), 7.04 (d, J = 8.0 Hz, 1H), 6.94 (d, J = 7.8 Hz, 1H), 4.81−4.77 (m, 1H), 4.69 (d, J = 1.1 Hz, 1H), 4.26 (dd, J = 11.9, 1.3 Hz, 2H), 4.21 (s, 5H), 4.01 (d, J = 11.4 Hz, 3H), 3.69 (d, J = 11.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 164.7, 141.4 (d, J_{pc} = 6.3 Hz), 137.6, 136.1, 134.1, 133.4, 130.6, 130.1, 129.3 (d, J_{pc} = 170.0 Hz), 86.8, 73.1 (d, J_{pc} = 4.5 Hz), 69.2, 68.4, 68.0, 67.9, 67.1, 53.6 (d, J_{pc} = 5.1 Hz), 52.9 (d, J_{pc} = 6.1 Hz). HRMS (ESI) m/z: $[M + K]^+$ calcd for $C_{28}H_{26}FeIO_6PK$ 710.9493; found 710.9475.

*(E)-Dimethyl[3-ferrocenyl-3-phenylmethylester-2-iodo-1-(1,1′-biphenyl-4-yl)propenyl]phosphonate ((E)-**3q**).* Light yellow solid; 103 mg, 85% yield; R_f = 0.40 (petroleum ether/ethyl acetate = 5:1); mp 120−121 °C. IR (neat, cm^{-1}) 2925, 2846, 1714, 1269, 1215, 1173, 1025, 839, 731.708. ^1H NMR (500 MHz, $CDCl_3$) δ 8.29 (d, J = 7.2 Hz, 2H), 7.74 (d, J = 2.6 Hz, 1H), 7.69−7.54 (m, 7H), 7.44 (t, J = 7.7 Hz, 2H), 7.36 (t, J = 7.4 Hz, 1H), 7.17 (d, J = 7.1 Hz, 1H), 7.08 (d, J = 6.8 Hz, 1H), 4.83 (s, 1H), 4.71 (s, 1H), 4.27 (dd, J = 8.8, 1.2 Hz, 2H), 4.23 (s, 5H), 4.04 (d, J = 11.4 Hz, 3H), 3.71 (d, J = 11.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 164.7, 141.9, 140.7, 140.4, 133.3, 130.0 (d, J_{pc} = 6.3 Hz), 129.5, 128.9, 128.8 (d, J_{pc} = 6.3 Hz), 128.6 (d, J_{pc} = 6.3 Hz), 127.5, 127.0 (d, J_{pc} = 6.3 Hz), 126.9, 86.9, 73.2, 69.2 (d, J_{pc} = 6.3 Hz), 68.4, 68.0, 67.8, 67.2, 53.6, 53.1. HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{34}H_{31}FeIO_5P$ 733.0298; found 733.0304.

*(E)-Dimethyl[3-ferrocenyl-3-phenylmethylester-2-iodo-1-(4-bromophenyl)propenyl]phosphonate ((E)-**3r**).* Light yellow solid; 92 mg, 76% yield; R_f = 0.48 (petroleum ether/ethyl acetate = 5:1); mp 175−177 °C. IR (neat, cm^{-1}) 2921, 2851, 1721, 1636, 1473, 1261, 1105, 1024, 904, 828, 711, 643. ^1H NMR (500 MHz, $CDCl_3$) δ 8.24 (d, J = 7.4 Hz, 2H), 7.68−7.59 (m, 2H), 7.53 (t, J = 7.7 Hz, 2H), 7.45 (dd, J = 11.4, 8.7 Hz, 2H), 6.95 (d, J = 7.9 Hz, 1H), 6.85 (d, J = 7.9 Hz, 1H), 4.77 (s, 1H), 4.66 (s, 1H), 4.23 (d, J = 10.7 Hz, 2H), 4.18 (s, 5H), 3.99 (d, J = 11.4 Hz, 3H), 3.66 (d, J = 11.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 164.7, 141.9, 133.4, 131.6, 130.9, 130.3, 130.0, 129.3 (d, J_{pc} = 167.5 Hz), 122.4, 86.7, 73.1 (d, J_{pc} = 3.8 Hz), 69.2, 68.4, 68.0, 67.9, 67.1, 53.6 (d, J_{pc} = 5.2 Hz), 53.0 (d, J_{pc} = 6.1 Hz). HRMS (ESI) m/z: $[M]^+$ calcd for $C_{28}H_{25}BrFeIO_5P$ 733.9013; found 733.9003.

*(E)-Dimethyl[3-ferrocenyl-3-phenylmethylester-2-iodo-1-(4-chlorophenyl)propenyl]phosphonate ((E)-**3s**).* Light yellow solid; 86 mg, 69% yield; R_f = 0.48 (petroleum ether/ethyl acetate = 5:1); mp 168−170 °C. IR (neat, cm^{-1}) 2921, 2851, 1720, 1635, 1473, 1460, 1261, 1104, 1026, 830, 711, 643. ^1H NMR (500 MHz, $CDCl_3$) δ 8.24 (d, J = 7.6 Hz, 2H), 7.64 (dd, J = 15.5, 4.5 Hz, 2H), 7.53 (t, J = 7.7 Hz, 2H), 7.30 (t, J = 9.8 Hz, 2H), 7.01 (d, J = 7.9 Hz, 1H), 6.92 (d, J = 7.9 Hz, 1H), 4.77 (s, 1H), 4.67 (s, 1H), 4.24 (d, J = 11.9 Hz, 2H), 4.19 (s, 5H), 3.99 (d, J = 11.4 Hz, 3H), 3.66 (d, J = 11.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 164.7, 141.4 (d, J_{pc} = 6.5 Hz), 137.8, 137.6, 137.4, 136.0, 134.1, 133.4, 130.6, 129.3 (d, J_{pc} = 170.0 Hz), 86.8, 73.1 (d, J_{pc} = 4.4 Hz), 69.2, 68.4, 68.1, 67.9, 67.1, 53.6 (d, J_{pc} = 5.3 Hz), 52.9 (d, J_{pc} = 6.1 Hz). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{28}H_{26}ClFeIO_5P$ 690.9596; found 690.9597.

*(E)-Dimethyl[3-ferrocenyl-3-phenylmethylester-2-iodo-1-ethylpropenyl]phosphonate ((E)-**3t**).* Pale yellow oil; 72 mg, 53% yield; R_f = 0.41 (petroleum ether/ethyl acetate = 5:1). IR (neat, cm^{-1}) 2922, 2852, 1719, 1648, 1518, 1458, 1265, 1107, 1026, 825, 756, 712. ^1H NMR (500 MHz, $CDCl_3$) δ 8.24 (d, J = 7.7 Hz, 2H), 7.62 (t, J = 7.0 Hz, 1H), 7.57−7.49 (m, 3H), 4.62 (d, J = 10.5 Hz, 2H), 4.21 (s, 1H), 4.16 (d, J = 6.5 Hz, 6H), 3.95 (d, J = 11.4 Hz, 3H), 3.82 (d, J = 11.2 Hz, 3H), 2.44 (dt, J = 14.8, 7.4 Hz, 2H), 1.01 (t, J = 7.3 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 164.4, 136.2, 135.0, 134.4, 133.3, 130.2, 129.3 (d, J_{pc} = 172.5 Hz), 87.2, 73.3 (d, J_{pc} = 5.4 Hz), 69.1, 68.2, 67.8, 67.6, 67.2, 53.3, 52.7 (d, J_{pc} = 5.5 Hz), 34.7 (d, J_{pc} = 8.4 Hz), 12.0. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{24}H_{26}FeIO_5P$ 607.9908; found 607.9907.

General Procedure for the Electrophilic Addition of Ferrocene-Containing Allenylphosphonates with Alcohols. To a solution of allenylphosphonate **1** (0.20 mmol), alcohol **4** (0.40 mmol), and *t*-BuOLi (0.20 mmol) in CH_2Cl_2 (3.0 mL) was added I_2 (0.20 mmol) under a N_2 atmosphere. The resulting mixture was stirred at room temperature for the indicated time. After completion of the reaction, the solvent was removed in a vacuum. The resulting residue was purified on a silica gel column (petroleum ether/EtOAc = 5:1) to provide the desired tetrasubstituted alkenylphosphonate products (*E*)-**5**.

*(E)-Dimethyl[3-ferrocenyl-3-ethoxyl-2-iodo-1-phenylpropenyl]-phosphonate ((E)-**5a**).* Light yellow solid; 70 mg, 62% yield; R_f = 0.43 (petroleum ether/ethyl acetate = 5:1); mp 129−130 °C. IR (neat, cm^{-1}) 2921, 2852, 1650, 1566, 1460, 1247, 1098, 1051, 1026, 822, 737, 702. ^1H NMR (500 MHz, $CDCl_3$) δ 7.43−7.31 (m, 3H), 7.13 (d, J = 7.2 Hz, 1H), 7.01 (s, 1H), 5.90 (d, J = 2.6 Hz, 1H), 4.56 (s, 1H), 4.47 (s, 1H), 4.26 (s, 5H), 4.20−4.13 (m, 2H), 3.78−3.64 (m, 8H), 1.42 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 144.1, 143.9, 143.2 (d, J_{pc} = 7.5 Hz), 138.3 (d, J_{pc} = 166.3 Hz), 128.8 (d, J_{pc} = 12.2 Hz), 128.4 (d, J_{pc} = 8.8 Hz), 128.0, 89.0, 77.8 (d, J_{pc} = 5.0 Hz), 69.1, 67.8, 67.3, 67.0, 64.2, 53.0 (d, J_{pc} = 3.8 Hz), 52.9 (d, J_{pc} = 3.8 Hz), 15.5. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{23}H_{26}FeIO_4P$ 579.9958; found 579.9954.

*(E)-Dimethyl[3-ferrocenyl-3-(2-bromoethoxyl)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**5b**).* Light yellow solid; 74 mg, 57% yield; R_f = 0.49 (petroleum ether/ethyl acetate = 5:1); mp 144−145 °C. IR (neat, cm^{-1}) 2920, 2849, 1646, 1461, 1245, 1106, 1050, 1026, 827, 701. ^1H NMR (500 MHz, $CDCl_3$) δ 7.38 (dd, J = 17.8, 5.5 Hz, 3H), 7.12 (d, J = 7.1 Hz, 1H), 7.01 (s, 1H), 6.02 (d, J = 2.5 Hz, 1H), 4.58 (s, 1H), 4.42 (s, 1H), 4.28 (s, 5H), 4.18 (dd, J = 10.0, 1.1 Hz, 2H), 4.05−3.92 (m, 2H), 3.72 (dd, J = 18.0, 11.2 Hz, 8H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 142.9 (d, J_{pc} = 7.4 Hz), 141.8, 139.2 (d, J_{pc} = 166.3 Hz), 128.7, 128.4, 128.2, 88.3, 78.3 (d, J_{pc} = 5.2 Hz), 69.2, 68.4, 68.0, 67.8, 67.4, 66.8, 63.4, 53.2 (d, J_{pc} = 6.3), 53.0 (d, J_{pc} = 6.3), 30.4. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{23}H_{25}BrFeIO_4P$ 657.9064; found 657.9058.

*(E)-Dimethyl[3-ferrocenyl-3-(4-iododibenzyloxyl)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**5c**).* Light yellow solid; 86 mg, 57% yield; R_f = 0.51 (petroleum ether/ethyl acetate = 5:1); mp 130−131 °C. IR (neat, cm^{-1}) 2949, 1572, 1480, 1244, 1181, 1052, 1023, 821, 774, 700. ^1H NMR (500 MHz, $CDCl_3$) δ 7.76 (d, J = 8.3 Hz, 2H), 7.38 (m, 5H), 7.11 (d, J = 7.3 Hz, 1H), 7.03 (s, 1H), 6.07 (d, J = 2.6 Hz, 1H), 4.67 (dd, J = 30.6, 11.7 Hz, 2H), 4.60−4.53 (m, 1H), 4.49−4.41 (m, 1H), 4.19 (s, 7H), 3.75 (d, J = 11.2 Hz, 3H), 3.67 (d, J = 11.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 143.0, 139.7, 138.0, 137.4, 130.0, 128.8, 128.6 (d, J_{pc} = 34.1 Hz), 128.2, 93.2, 88.6, 78.3 (d, J_{pc} = 5.0 Hz), 70.0, 69.1, 67.9 (d, J_{pc} = 13.9 Hz), 67.3, 66.8, 53.1 (d, J_{pc} = 6.3 Hz), 53.0 (d, J_{pc} = 6.3 Hz). HRMS (ESI) m/z: $[M]^+$calcd for $C_{28}H_{27}FeI_2O_4P$ 767.9081; found 767.9081.

*(E)-Dimethyl[3-ferrocenyl-3-(4-oxy-1-butyne)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**5d**).* Light yellow solid; 75 mg, 63% yield; R_f = 0.53 (petroleum ether/ethyl acetate = 5:1); mp 120−121 °C. IR (neat, cm^{-1}) 2921, 2853, 1635, 1575, 1246, 1101, 1051, 1026, 826. 737, 703, 642. ^1H NMR (500 MHz, $CDCl_3$) δ 7.44−7.32

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(m, 3H), 7.13 (d, J = 7.3 Hz, 1H), 7.01 (s, 1H), 5.96 (d, J = 2.6 Hz, 1H), 4.57 (s, 1H), 4.46 (d, J = 0.9 Hz, 1H), 4.28 (s, 5H), 4.17 (dd, J = 10.3, 1.2 Hz, 2H), 3.83−3.75 (m, 2H), 3.73 (dd, J = 11.2, 0.9 Hz, 6H), 2.77−2.61 (m, 2H), 2.07 (s, 1H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 143.1, 142.7, 138.9 (d, J_{pc} = 167.5 Hz), 128.8, 128.4, 128.1, 88.6, 81.5, 78.1 (d, J_{pc} = 5.1 Hz), 69.5, 69.2, 67.8 (d, J_{pc} = 16.9 Hz), 67.3, 66.9, 66.6, 53.0 (t, J_{pc} = 6.3 Hz), 20.0. HRMS (ESI) m/z: $[M]^+$calcd for $C_{25}H_{26}FeIO_4P$ 603.9958; found 603.9956.

*(E)-Dimethyl[3-ferrocenyl-3-(2-methylheptyloxy)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**5e**).* Light yellow solid; 77 mg, 59% yield; R_f = 0.39 (petroleum ether/ethyl acetate = 5:1); mp 86−87 °C. IR (neat, cm^{-1}) 2923, 2854, 1650, 1571, 1460, 1247, 1052, 1027, 821, 699. ^1H NMR (500 MHz, $CDCl_3$) δ 7.38 (d, J = 23.6 Hz, 3H), 7.12 (d, J = 6.4 Hz, 1H), 7.02 (s, 1H), 5.86 (s, 1H), 4.56 (s, 1H), 4.44 (s, 1H), 4.25 (s, 5H), 4.16 (s, 2H), 3.72 (dd, J = 11.0, 7.5 Hz, 6H), 3.53−3.40 (m, 2H), 1.90 (d, J = 5.7 Hz, 1H), 1.32 (d, J = 38.9 Hz, 8H), 1.10 (d, J = 6.5 Hz, 3H), 0.92 (d, J = 6.6 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 144.0, 143.2 (d, J_{pc} = 7.5 Hz), 138.3, (d, J_{pc} = 171.3 Hz), 128.8 (d, J_{pc} = 14.6 Hz), 128.3 (d, J_{pc} = 7.3 Hz), 128.0, 89.3, 78.2, 73.9, 69.1, 67.9, 67.1, 66.7, 52.9 (d, J_{pc} = 3.8 Hz), 52.9 (d, J_{pc} = 3.8 Hz), 33.8 (d, J_{pc} = 4.5 Hz), 32.3 (d, J_{pc} = 6.1 Hz), 26.9 (d, J_{pc} = 6.0 Hz), 22.8, 17.6, 14.2. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{29}H_{38}FeIO_4P$ 664.0897; found 664.0892.

*(E)-Dimethyl[3-ferrocenyl-3-cyclohexanoxyl-2-iodo-1-phenylpropenyl]phosphonate ((E)-**5f**).* Light yellow solid; 62 mg, 50% yield; R_f = 0.43 (petroleum ether/ethyl acetate = 5:1); mp 119−120 °C. IR (neat, cm^{-1}) ^1H NMR (500 MHz, $CDCl_3$) δ 7.45−7.32 (m, 3H), 7.12 (d, J = 7.3 Hz, 1H), 7.03 (s, 1H), 6.13 (d, J = 2.5 Hz, 1H), 4.56 (s, 1H), 4.41 (s, 1H), 4.26 (s, 5H), 4.15 (s, 2H), 3.73 (dd, J = 11.2, 5.7 Hz, 6H), 3.61−3.47 (m, 1H), 1.47 (m, 11H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 145.3, 145.2, 143.3 (d, J_{pc} = 7.5 Hz), 137.5 (d, J_{pc} = 166.3 Hz), 129.0, 128.7, 128.4 (d, J_{pc} = 8.8 Hz), 128.0, 89.6, 75.1 (d, J_{pc} = 5.0 Hz), 69.1, 67.9 (d, J_{pc} = 9.1 Hz), 67.0, 66.6, 52.9 (d, J_{pc} = 6.3 Hz), 52.8 (d, J_{pc} = 7.5 Hz), 33.8, 31.9, 26.0, 24.4, 24.2. HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{27}H_{32}FeIO_4PNa$ 657.0323; found 657.0298.

*Dimethyl[3-ferrocenyl-3-tert-butoxy-2-iodo-1-phenylpropenyl]-phosphonate ((E)-**5g**).* Light yellow solid; 17 mg, 14% yield; R_f = 0.38 (petroleum ether/ethyl acetate = 5:1); mp 142−143 °C. IR (neat, cm^{-1}) 2920, 2851, 1731, 1640, 1460, 1459, 1052, 1245, 1188, 1054, 1030, 899, 822, 701. ^1H NMR (500 MHz, $CDCl_3$) δ 7.71 (d, J = 8.0 Hz, 1H), 7.37 (d, J = 7.0 Hz, 2H), 7.14 (d, J = 7.9 Hz, 1H), 7.07 (d, J = 6.7 Hz, 1H), 6.36 (s, 1H), 4.68 (s, 1H), 4.52 (s, 1H), 4.29 (s, 5H), 4.16 (s, 2H), 3.79 (d, J = 11.2 Hz, 3H), 3.68 (d, J = 11.1 Hz, 3H), 1.48 (s, 9H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 143.0 (d, J = 7.5 Hz), 142.5, 142.3, 139.7, 138.0, 137.4, 130.0, 128.7 (d, J = 3.7 Hz), 128.4, 128.2, 93.2, 88.6, 78.3 (d, J = 5.1 Hz), 70.0, 69.1 (d, J = 9.5 Hz), 67.9 (d, J = 13.6 Hz), 67.3, 66.8, 53.0 (dd, J = 13.7, 6.5 Hz), 29.0. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{25}H_{30}FeIO_4P$ 608.0271; found 608.0271.

*(E)-Dimethyl[3-ferrocenyl-3-ethoxyl-2-iodo-1-(3,5-dimethylphenyl)propenyl]phosphonate ((E)-**5h**).* Light yellow solid; 74 mg, 66% yield; R_f = 0.42 (petroleum ether/ethyl acetate = 5:1); mp 149−150 °C. IR (neat, cm^{-1}) 2921, 2852, 1637, 1464, 1244, 1100, 1055, 1025, 822, 751, 701. ^1H NMR (500 MHz, $CDCl_3$) δ 6.98 (s, 1H), 6.72 (s, 1H), 6.61 (s, 1H), 5.85 (d, J = 2.5 Hz, 1H), 4.57 (s, 1H), 4.48 (s, 1H), 4.26 (s, 5H), 4.17 (d, J = 10.4 Hz, 2H), 3.72 (dd, J = 12.9, 11.3 Hz, 8H), 2.32 (d, J = 28.4 Hz, 6H), 1.42 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 143.5, 143.3, 142.9 (d, J_{pc} = 7.4 Hz), 138.5 (d, J_{pc} = 165.0 Hz), 137.8, 129.7, 126.3 (d, J_{pc} = 12.2 Hz), 89.1, 77.9 (d, J_{pc} = 5.1 Hz), 69.1, 67.9 (d, J_{pc} = 7.5 Hz), 67.3, 67.1, 64.1, 52.9 (d, J_{pc} = 6.3 Hz), 21.4 (d, J_{pc} = 10.2 Hz), 15.5. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{25}H_{30}FeIO_4P$ 608.0271; found 608.0274.

*(E)-Dimethyl[3-ferrocenyl-3-ethoxyl-2-iodo-1-(1,1′-biphenyl-4-yl)propenyl]phosphonate ((E)-**5i**).* Light yellow solid; 63 mg, 58% yield; R_f = 0.37 (petroleum ether/ethyl acetate = 5:1); mp 146−147 °C. IR (neat, cm^{-1}) 2921, 2852, 1642, 1470, 1247, 1099, 1028, 824, 749, 699. ^1H NMR (500 MHz, $CDCl_3$) δ 7.68−7.56 (m, 4H), 7.46 (t, J = 7.6 Hz, 2H), 7.37 (t, J = 7.3 Hz, 1H), 7.20 (d, J = 7.2 Hz, 1H), 7.09 (d, J = 7.2 Hz, 1H), 5.93 (d, J = 2.5 Hz, 1H), 4.58 (s, 1H), 4.49 (s, 1H), 4.27 (s, 5H), 4.18 (d, J = 10.9 Hz, 2H), 3.76 (dd, J = 11.2, 8.6 Hz, 8H), 1.44 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 144.3, 144.1, 142.1 (d, J_{pc} = 7.5 Hz), 140.7, 140.3, 138.0 (d, J_{pc} = 167.5 Hz), 129.2 (d, J_{pc} = 19.2 Hz), 128.8, 127.5, 127.1, 89.0, 77.8 (d, J_{pc} = 5.0 Hz), 69.1, 67.8, 67.3, 67.1, 64.2, 53.0 (d, J_{pc} = 3.8 Hz), 52.9 (d, J_{pc} = 3.8 Hz), 29.7, 15.6. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{29}H_{30}FeIO_4P$ 656.0271; found 656.0274.

*(E)-Dimethyl[3-ferrocenyl-3-ethoxyl-2-iodo-1-(4-bromophenyl)-propenyl]phosphonate ((E)-**5j**).* Light yellow solid; 58 mg, 54% yield; R_f = 0.47 (petroleum ether/ethyl acetate = 5:1); mp 140−141 °C. IR (neat, cm^{-1}) 2921, 2852, 1637, 1476, 1247, 1050, 1023, 826, 734, 703. ^1H NMR (500 MHz, $CDCl_3$) δ 7.53 (d, J = 8.1 Hz, 1H), 7.48 (d, J = 8.1 Hz, 1H), 7.00 (d, J = 7.9 Hz, 1H), 6.88 (d, J = 7.9 Hz, 1H), 5.88 (d, J = 2.5 Hz, 1H), 4.55 (s, 1H), 4.44 (s, 1H), 4.25 (s, 5H), 4.16 (d, J = 14.1 Hz, 2H), 3.73 (dd, J = 11.2, 5.7 Hz, 6H), 3.71−3.61 (m, 2H), 1.41 (t, J = 7.0 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 144.8, 144.7, 142.0 (d, J_{pc} = 7.6 Hz), 137.2 (d, J_{pc} = 168.8 Hz), 131.6, 130.6 (d, J_{pc} = 18.2 Hz), 122.3, 88.8, 77.7 (d, J_{pc} = 4.9 Hz), 69.1, 67.8, 67.8, 67.4, 67.0, 64.2, 53.0 (d, J_{pc} = 4.8 Hz), 15.5. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{23}H_{25}BrFeIO_4P$ 657.9064; found 657.9065.

General Procedure for the Electrophilic Addition of Ferrocene-Containing Allenylphosphonates with Phenols. To a solution of allenylphosphonate **1** (0.20 mmol), phenol **6** (0.40 mmol), and *t*-BuOLi (0.20 mmol) in CH_2Cl_2 (3.0 mL) was added I_2 (0.20 mmol) under a N_2 atmosphere. The resulting mixture was stirred at room temperature for the indicated time. After completion of the reaction, the solvent was removed in a vacuum. The resulting residue was purified on a silica gel column (petroleum ether/EtOAc = 5:1) to provide the desired tetrasubstituted alkenylphosphonate products (*E*)-**7**.

*(E)-Dimethyl[3-ferrocenyl-3-phenoxyl-2-iodo-1-phenylpropenyl]-phosphonate ((E)-**7a**).* Light yellow solid; 75 mg, 61% yield; R_f = 0.42 (petroleum ether/ethyl acetate = 5:1); mp 177−178 °C. IR (neat, cm^{-1}) 2920, 2850, 1643, 1467, 1237, 1032, 824, 737, 700, 644. ^1H NMR (500 MHz, $CDCl_3$) δ 7.40 (q, J = 8.1 Hz, 4H), 7.33 (s, 3H), 7.15 (s, 1H), 7.09 (t, J = 6.6 Hz, 1H), 7.00 (s, 1H), 6.89 (s, 1H), 4.76 (s, 1H), 4.63 (s, 1H), 4.26, (s, 5H), 4.23 (s, 2H), 3.74 (d, J = 11.2 Hz, 3H), 3.68 (d, J = 11.3 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 157.1, 143.0 (d, J_{pc} = 6.6 Hz), 141.7, 141.5, 137.9 (d, J_{pc} = 165.0 Hz), 129.5, 128.8, 128.3, 128.1, 121.9, 117.5, 88.3, 75.8 (d, J_{pc} = 4.3 Hz), 69.2, 68.1, 67.7 (d, J_{pc} = 2.7 Hz), 67.3, 53.0 (d, J_{pc} = 5.9 Hz). HRMS (ESI) m/z: $[M]^+$ calcd for $C_{27}H_{26}FeIO_4P$ 627.9958; found 627.9959.

*(E)-Dimethyl[3-ferrocenyl-3-(4-methylphenoxyl)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**7b**).* Light yellow solid; 76 mg, 59% yield; R_f = 0.52 (petroleum ether/ethyl acetate = 5:1); mp 122−123 °C. IR (neat, cm^{-1}) 2922, 2848, 1504, 1232, 1027, 827, 731, 691. ^1H NMR (500 MHz, $CDCl_3$) δ 7.32−7.27 (m, 3H), 7.22 (d, J = 8.5 Hz, 2H), 7.16 (d, J = 8.5 Hz, 2H), 7.04 (d, J = 2.7 Hz, 1H), 6.96 (s, 1H), 6.87 (s, 1H), 4.71 (d, J = 0.9 Hz, 1H), 4.58 (d, J = 1.0 Hz, 1H), 4.22 (s, 5H), 4.21−4.15 (m, 2H), 3.70 (d, J = 11.2 Hz, 3H), 3.65 (d, J = 11.3 Hz, 3H), 2.33 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 154.9, 143.0 (d, J = 6.8 Hz), 142.1, 141.9, 137.0, 131.0, 129.9, 128.7, 128.2, 128.0, 117.2, 88.4, 75.9 (d, J = 4.5 Hz), 69.1, 67.9, 67.6 (d, J = 11.8 Hz), 67.2, 52.9 (dd, J = 6.0, 2.6 Hz), 20.6. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{28}H_{28}FeIO_4P$ 642.0114; found 642.0100.

*(E)-Dimethyl[3-ferrocenyl-3-(4-tert-butylphenoxyl)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**7c**).* Pale yellow oil; 31 mg, 23% yield, R_f = 0.38 (petroleum ether/ethyl acetate = 5:1). IR (neat, cm^{-1}) 2922, 2853, 1692, 1642, 1506, 1267, 1032, 829, 755. ^1H NMR (500 MHz, $CDCl_3$) δ 7.37 (d, J = 8.5 Hz, 2H), 7.30 (s, 3H), 7.24 (s, 2H), 7.02 (s, 1H), 6.97 (s, 1H), 6.88 (s, 1H), 4.71 (s, 1H), 4.56 (s, 1H), 4.22 (s, 5H), 4.18 (s, 2H), 3.69 (d, J = 11.2 Hz, 3H), 3.59 (d, J = 11.3 Hz, 3H), 1.33 (s, 9H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 154.9, 144.6, 143.0 (d, J = 6.9 Hz), 142.1, 142.0, 138.2, 136.9, 128.7 (d, J = 20.7 Hz), 128.3, 128.0, 126.2, 117.1, 88.5, 76.4 (d, J = 4.5 Hz), 69.1, 67.9, 67.6 (d, J = 17.1 Hz), 67.2, 52.9 (dd, J = 9.1, 6.0 Hz), 31.6, 29.7. HRMS (ESI) m/z: $[M]^+$calcd for $C_{31}H_{34}FeIO_4P$ 684.0583; found 684.0580.

*(E)-Dimethyl[3-ferrocenyl-3-(4-bromophenoxyl)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**7e**).* Light yellow solid; 69 mg,

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50% yield; R_f = 0.52 (petroleum ether/ethyl acetate = 5:1); mp 147−148 °C. IR (neat, cm^{-1}) 2921, 2852, 1725, 1634, 1580, 1486, 1278, 1263, 1032, 824, 700. ^1H NMR (500 MHz, $CDCl_3$) δ 7.49 (d, J = 8.6 Hz, 2H), 7.34 (s, 3H), 7.26 (d, J = 8.6 Hz, 2H), 7.09 (d, J = 1.8 Hz, 1H), 6.98 (s, 1H), 6.90 (s, 1H), 4.71 (s, 1H), 4.60 (s, 1H), 4.26 (s, 2H), 4.23 (s, 5H), 3.75 (d, J = 11.3 Hz, 3H), 3.69 (d, J = 11.2 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 142.8 (d, J_{pc} = 6.6 Hz), 140.7 (d, J_{pc} = 22.8 Hz), 138.3 (d, J_{pc} = 165.0 Hz), 132.4, 128.7, 128.4, 128.3 (d, J_{pc} = 20.4 Hz), 119.1, 117.3, 114.1, 87.8, 75.8 (d, J_{pc} = 4.4 Hz), 69.2, 68.1, 67.8, 67.6, 67.3, 53.2, 53.1 (d, J_{pc} = 2.5 Hz). HRMS (ESI) m/z: $[M]^+$ calcd for $C_{27}H_{25}BrFeIO_4P$ 705.9064; found 705.9062.

*(E)-Dimethyl[3-ferrocenyl-3-(3-fluorophenoxyl)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**7f**).* Light yellow solid; 66 mg, 52% yield; R_f = 0.43 (petroleum ether/ethyl acetate = 5:1); mp 155−156 °C. IR (neat, cm^{-1}) 2923, 2853, 1603, 1497, 1207, 1051, 838, 737, 701. ^1H NMR (500 MHz, $CDCl_3$) δ 9.75 (s, 1H), 7.90−7.80 (m, 1H), 7.59−7.46 (m, 4H), 7.30 (s, 1H), 7.15 (s, 1H), 6.94 (s, 2H), 6.77 (d, J = 5.8 Hz, 1H), 4.55 (s, 1H), 4.36 (d, J = 11.3 Hz, 5H), 4.26 (d, J = 19.7 Hz, 2H), 4.03 (d, J = 11.2 Hz, 3H), 3.79 (d, J = 11.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 164.0, 157.2, 143.6, 132.0, 129.6, 128.7 (dd, J_{pc} = 26.8, 13.2 Hz), 122.1, 106.3 (d, J_{pc} = 21.3 Hz), 105.2 (d, J_{pc} = 22.5 Hz), 91.4, 69.8, 69.3, 68.9, 68.0, 67.4, 53.7 (d, J_{pc} = 19.0 Hz), 45.2. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{27}H_{25}FFeIO_4P$ 645.9863; found 645.9866.

*(E)-Dimethyl[3-ferrocenyl-3-(3-methyl-4-chlorophenoxyl)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**7g**).* Light yellow solid; 64 mg, 48% yield; R_f = 0.55 (petroleum ether/ethyl acetate = 5:1); mp 128−129 °C. IR (neat, cm^{-1}) 2921, 2851, 2362, 1651, 1568, 1475, 1286, 1247, 1170, 1031, 820, 699. ^1H NMR (500 MHz, $CDCl_3$) δ 7.36−7.30 (m, 4H), 7.23 (d, J = 2.1 Hz, 1H), 7.15 (dd, J = 8.7, 2.4 Hz, 1H), 7.07 (s, 1H), 6.98 (s, 1H), 6.90 (s, 1H), 4.72 (s, 1H), 4.62 (s, 1H), 4.24 (s, 7H), 4.22 (s, 2H), 3.76 (d, J = 11.3 Hz, 3H), 3.71 (d, J = 11.2 Hz, 3H), 2.44 (s, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 155.5, 142.9 (d, J_{pc} = 6.6 Hz), 141.1 (d, J_{pc} = 22.5 Hz), 138.2 (d, J_{pc} = 166.3 Hz), 137.1, 129.7, 128.7, 128.3, 128.2, 126.8, 119.5, 115.7, 88.0, 75.6 (d, J_{pc} = 4.4 Hz), 69.2, 68.1, 67.7 (d, J_{pc} = 15.6 Hz), 67.3, 53.0 (t, J_{pc} = 6.3 Hz), 20.4. HRMS (ESI) m/z: $[M]^+$ calcd for $C_{28}H_{27}ClFeIO_4P$ 675.9724; found 675.9732.

*(E)-Dimethyl[3-ferrocenyl-3-(4-phenylphenoxyl)-2-iodo-1-phenylpropenyl]phosphonate ((E)-**7i**).* Light yellow solid; 55 mg, 40% yield; R_f = 0.38 (petroleum ether/ethyl acetate = 5:1); mp 118−119 °C. IR (neat, cm^{-1}) 2922, 2851, 1606, 1518, 1484, 1229, 1039, 833, 762, 698. ^1H NMR (500 MHz, $CDCl_3$) δ 7.63 (d, J = 8.4 Hz, 4H), 7.49−7.41 (m, 4H), 7.34 (d, J = 11.5 Hz, 4H), 7.16 (s, 1H), 7.00 (s, 1H), 6.92 (s, 1H), 4.77 (s, 1H), 4.64 (s, 1H), 4.27 (s, 5H), 4.24 (s, 2H), 3.76 (d, J = 11.3 Hz, 3H), 3.70 (d, J = 11.3 Hz, 3H). $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 156.7, 155.6, 142.9 (d, J_{pc} = 6.8 Hz), 141.8, 141.6, 140.8 (d, J_{pc} = 9.9 Hz), 137.7 (d, J_{pc} = 166.3 Hz), 134.9, 133.7, 128.8 (d, J_{pc} = 7.8 Hz), 128.3 (d, J_{pc} = 19.1 Hz), 126.8, 126.7, 126.6, 117.6, 115.7, 88.2, 76.0 (d, J_{pc} = 4.4 Hz), 69.2, 68.0, 67.7, 67.3, 53.2 (d, J_{pc} = 2.5 Hz), 53.1 (d, J_{pc} = 2.5 Hz). HRMS (ESI) m/z: $[M]^+$ calcd for $C_{33}H_{30}FeIO_4P$ 704.0271; found 704.0270.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.0c00762.

X-ray analysis data of single-crystal **3a**, and copies of ^1H and $^{13}C\{^1H\}$ NMR spectra for all new compounds (PDF)

Crystallographic data file for **3a** (CIF)

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Notes

The authors declare no competing financial interest.

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